

PYTHON WEATHER BONANZA

Detailed Lesson Guide • Grade 4

Subject	Computer Science / Coding
Grade Level	Grade 4 (Ages 9–10)
Duration	60 minutes
Platform	Python 3 (any IDE or online compiler)
Curriculum Unit	Introduction to Programming
Source Material	Planrs Academy – Python Weather Bonanza

"Every big programmer started just like you!"

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1 Lesson Overview & Learning Objectives

Story Hook

Our lesson opens with Priya, a curious girl in Jaipur, who hears her mother announce it's 42°C outside on a scorching summer morning. Priya wonders — *can she check the weather herself, and even build a program to analyse it?* This relatable scenario grounds abstract coding concepts in a real-world Indian context familiar to Grade 4 students.

Learning Objectives

By the end of this lesson, students will be able to:

- [>] Define what Python is and name at least two real-world applications.
- [>] Create and assign values to variables using correct Python syntax.
- [>] Use the `print()` function to display text and variable values.
- [>] Write a basic if-else statement that makes a decision based on temperature.
- [>] Explain the IPO (Input - Process - Output) model of computing.
- [>] Build and run a working Python Weather Analyser program.

Materials Required

Item	Details
Computer / Tablet	One per student or per pair
Python Environment	IDLE, Thonny, Replit, or any online Python compiler
This Lesson Sheet	Printed or on-screen reference
Whiteboard / Projector	For teacher demonstration

Lesson Flow (60 minutes)

Time	Activity	Method
0–5 min	Story hook — Priya in Jaipur	Teacher narration
5–10 min	What is Python? Real-world examples	Discussion
10–20 min	Variables: concept & syntax	Teach + Try
20–30 min	<code>print()</code> function & output	Teach + Try
30–45 min	If-Else decisions + IPO model	Teach + Code-Along
45–55 min	Starter Code activity — fill the blanks	Independent/Pairs
55–60 min	Share, celebrate, extension challenge	Whole Class

2 Background: What is Python?

Teacher Introduction

Python is a **programming language** — a special set of instructions that tells a computer exactly what to do, step by step. Just like Amma's sambar recipe lists ingredients and steps in order, Python code is a recipe for the computer.

Python was created by **Guido van Rossum** in 1991. It is named after the British comedy show *Monty Python's Flying Circus* — NOT after the snake! Today it is one of the world's most popular programming languages.

Where is Python Used in India?

Organisation	How they use Python
ISRO	Satellite data analysis & space mission simulations
Zomato / Swiggy	Processing millions of food orders & delivery routing
IRCTC	Managing train bookings & passenger data
Google India	Search algorithms, AI, and data analytics
Netflix India	Recommendation engines & streaming quality control

Why Python is Perfect for Beginners

[YES] Reads almost like English — easy to understand

[YES] Simple syntax — no complicated curly braces or semicolons needed

[YES] Huge community — millions of people use it and can help

[YES] Free to use — anyone can download and run Python for free

[YES] Runs everywhere — Windows, Mac, Linux, or even in a browser

Discussion Question for Students

"Can you think of an app on your phone that might use Python? What problem does it solve?"

3 Core Concept: Variables

What is a Variable?

A **variable** is like a labelled box (or container) that holds a piece of information. Think of Amma's *masala dabba* — the spice box has separate compartments labelled 'Turmeric', 'Cumin Seeds', and 'Chilli'. Each compartment stores one type of spice. In Python, each variable stores one piece of data!

Analogy Table

Masala Dabba	Python Variable
Compartment label	Variable name (e.g. city)
Spice inside	Variable value (e.g. "Mumbai")
Putting spice in	Assignment with = (e.g. city = "Mumbai")
Reading the label	Using the variable in print()

Variable Syntax Rules

Rule 1	Start with a letter or underscore — NEVER a number
Rule 2	Use underscores for spaces (city_name, NOT city name)
Rule 3	= means STORE, not 'equals' like in maths
Rule 4	Text values need quotation marks: city = "Delhi"
Rule 5	Number values do NOT need quotes: temperature = 45

Example Code — Indian City Weather Variables

```
# Weather variables for Indian cities
city1 = "Mumbai"
temperature1 = 38

city2 = "Delhi"
temperature2 = 45

city3 = "Shimla"
temperature3 = 18
```

Student Activity: Fill in the Blanks

Write variables for YOUR favourite city:

```
city = "_____"  
temperature = _____
```

4 Core Concept: The print() Function

What is print()?

The **print()** function tells Python to display something on the screen. Think of it like a TV weather report — the anchor reads the data out loud so everyone can see it. In Python, print() reads your data out to the screen!

How to Use print()

What to print	Example Code	Output on Screen
Plain text	<code>print("Hello, Priya!")</code>	Hello, Priya!
A number variable	<code>print(temperature)</code>	38
Text + variable	<code>print("Temp:", temperature)</code>	Temp: 38
Multiple items	<code>print("City:", city, "Temp:", temperature)</code>	City: Mumbai Temp: 38
Text with units	<code>print("Temperature:", temperature, "degrees Celsius")</code>	Temperature: 38 degrees Celsius

Common Mistakes to Avoid

Status	Code	Explanation
WRONG	<code>print(Hello)</code>	Strings need quotes — Python doesn't know Hello is text
WRONG	<code>Print("Hello")</code>	Python is case-sensitive — use lowercase print
WRONG	<code>print "Hello"</code>	Always use brackets: <code>print("Hello")</code>
RIGHT	<code>print("Hello, World!")</code>	Correct!

Practice — Weather Report Output

Type and run this complete code:



```
city = "Jaipur"
temperature = 42

print("=== WEATHER REPORT ===")
print("City:", city)
print("Temperature:", temperature, "degrees Celsius")
```


5 Core Concept: If-Else Decisions

What is an If-Else Statement?

An **if-else** statement lets a program make a **decision**. Just like your parents decide: *'IF it's very hot outside, wear a cap and carry water. ELSE, you can play normally!'* — your program can make the same kind of choices.

Real-Life to Code Comparison

Real Life Decision	Python Code
IF temperature > 35 C Drink lots of water! ELSE Enjoy the weather!	<pre>if temperature > 35: print("Hot day! Drink plenty of water.") else: print("Cool day! Enjoy the weather.")</pre>

Syntax Rules for If-Else

1. Always write if in **lowercase**
2. Put a **colon :** after the condition — it is mandatory
3. **Indent** the lines inside each block (4 spaces or 1 Tab)
4. Use **==** to compare values, **=** to store values
5. The else block is optional — use it when you need a 'backup' action

Extended Example — Three Temperature Levels

We can use **elif** (else-if) to handle more than two outcomes:

```
temperature = 42

if temperature > 40:
    print("Extreme heat! Stay indoors and drink water.")
elif temperature > 30:
    print("Hot day! Carry an umbrella and wear light clothes.")
else:
    print("Pleasant weather! Great day to go outside.")
```

Indian City Temperature Decision Table

City	Temp	Program Output
Delhi (summer)	45 C	Extreme heat! Stay indoors and drink water.

Jaipur	42 C	Extreme heat! Stay indoors and drink water.
Mumbai	38 C	Hot day! Carry an umbrella and wear light clothes.
Chennai (monsoon)	28 C	Pleasant weather! Great day to go outside.
Shimla (summer)	18 C	Pleasant weather! Great day to go outside.

6 The IPO Model

What is the IPO Model?

All computer programs follow a simple pattern called the **IPO Model**: **Input** → **Process** → **Output**. Think of a juice stall — you give fruit (Input), the juicer blends it (Process), and you get yummy juice (Output). Every program, no matter how simple or complex, follows this same pattern.

INPUT	PROCESS	OUTPUT
Data that goes INTO the program Fruit (in)	What the computer DOES with the data Juicer blends it	The RESULT shown to the user Juice! (out)

IPO in Our Weather Analyser

IPO Stage	In Our Program
[IN] Input	<code>city = "Jaipur"</code> <code>temperature = 42</code>
[->] Process	<code>if temperature > 35:</code> <code># decide which message to show</code> <code>elif temperature > 30:</code> <code># different message</code>
[OUT] Output	<code>print("Weather in", city)</code> <code>print("Temperature:", temperature, "degrees")</code>

7 Hands-On Coding Activity

Instructions

- Step 1:** Open your Python editor (IDLE / Thonny / Replit / Google Colab).
- Step 2:** Create a new file and save it as `weather_analyser.py`
- Step 3:** Type the starter code below — do NOT copy-paste, type it yourself!
- Step 4:** Fill in the blanks (shown as `_____`) with your own values.
- Step 5:** Run the program and check the output.
- Step 6:** Try changing the temperature and see if the message changes.

Starter Code — Fill in the Blanks

```
# =====  
# Python Weather Analyser  
# My name: _____  
# Date: _____  
# =====  
  
# --- City 1 ---  
city1 = "_____" # enter a city name  
temperature1 = ____ # enter a temperature  
  
print("Weather in", city1)  
print("Temperature:", temperature1, "degrees")  
  
if temperature1 > 35:  
    print("_____  
else:  
    print("_____  
  
print() # blank line  
  
# --- City 2 ---  
city2 = "_____"  
temperature2 = ____  
  
print("Weather in", city2)  
print("Temperature:", temperature2, "degrees")  
  
if temperature2 > 35:  
    print("_____  
else:  
    print("_____")
```

Write Your Expected Output Here

8 Extension Challenge

Ready for More?

If you finish the main activity early, try these extension challenges. Each one builds on the skills you've learned today.

Challenge 1 — Third City

Add a third city (Bengaluru, Kolkata, Hyderabad, or your own city!) to your program. Use `city3`, `temperature3` and the same print / if-else pattern.

```
city3 = "Bengaluru"
temperature3 = 25
print("Weather in", city3)
# ... add your if-else here ...
```

Challenge 2 — Three Levels

Instead of just hot / cool, write an if-elif-else that handles THREE levels: Extreme Heat (>40 C), Hot (>30 C), and Cool (30 C or below).

```
if temperature > 40:
    print("Extreme heat! Stay indoors.")
elif temperature > 30:
    print("Hot day! Wear light clothes.")
else:
    print("Cool and pleasant!")
```

Challenge 3 — Seasons

Can you add a season variable? Decide whether it's Summer, Monsoon, or Winter based on the temperature. Print the season along with the weather advice.

```
if temperature > 35:
    season = "Summer"
elif temperature > 20:
    season = "Monsoon"
else:
    season = "Winter"
print("Season:", season)
```

9 Assessment & Reflection

Check Your Understanding — Quiz

#	Question	Answer
Q1	What is a variable in Python?	A labelled box that stores a value (data).
Q2	What does the = sign do in Python?	It stores a value in a variable — it does NOT mean 'equals'.
Q3	What do you need around text values in Python?	Quotation marks " " or ' '
Q4	What does print() do?	It displays text or variable values on the screen.
Q5	What two symbols follow the condition in an if statement?	Colon : and then an indented new line.
Q6	What does IPO stand for?	Input → Process → Output.
Q7	Name two Indian organisations that use Python.	Any two of: ISRO, Zomato, Swiggy, IRCTC, etc.
Q8	In the code <code>if temperature > 35:</code> what does <code>></code> mean?	Greater than — the condition is true if temperature is more than 35.

Reflection — Exit Ticket

Ask students to answer these three questions before leaving:

[+] **One thing I learned today:**

[?] **One thing I found tricky:**

[>>] **One way I could use Python in real life:**

10 Quick Reference Card

Cut out and keep this handy while coding!

Variables

<code>city = "Mumbai"</code>	Store text in a variable
<code>temperature = 38</code>	Store a number in a variable
<code>city_name = ...</code>	Use underscore for spaces

print()

<code>print("Hello")</code>	Print text
<code>print(temperature)</code>	Print a variable
<code>print("Temp:", temperature, "°C")</code>	Print multiple items

If-Else

<code>if condition:</code>	Start a decision block
<code>print(...)</code>	Indent 4 spaces inside block
<code>elif another_condition:</code>	Add more conditions
<code>else:</code>	Default when nothing matches

Comparison Operators

<code>></code>	greater than
<code><</code>	less than
<code>>=</code>	greater than or equal to
<code><=</code>	less than or equal to
<code>==</code>	equal to (compare)
<code>!=</code>	not equal to

Congratulations — You Are Now a Python Coder!

Every big programmer started exactly like you — with one line of code. Keep exploring, keep building, and remember: coding is creative and fun!